Gagan Bansal

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Summary

Final-year CS student (CGPA 9.41, graduating June 2026) with R&D internship at Quark Software. Hands-on experience in backend systems, applied AI development, and cross-platform C++ porting.

Technical Skills

Languages: C++, Java, Python(basic)
Web and Backend: React.js, TailwindCSS, Node.js, Express.js, Apache Kafka, RESTful APIs
Databases: MongoDB, SQL, PostgreSQL
GenAI: LLM APIs, RAG Pipelines, Pinecone, Agentic AI, MCP, LangChain
Tools and Platforms: Figma, Git, Linux, macOS, Claude Code

Experience

Research and Development Intern, *Quark Software Inc.* Jan 2026 – Present *Chandigarh*

- Ported and modernized a legacy codebase to ensure compatibility with the latest macOS versions, supporting both ARM64 (Apple Silicon) and x86-64 architectures.
- Resolved complex cross-platform build and dependency issues involving OpenSSL and Axis2, ensuring stable builds across ARM64 and x86 architectures.
- Debugged and fixed low-level compatibility issues arising from architecture differences and deprecated system APIs.

Projects

oForge - Engineering Workspace Platform demo
React.js, Node.js, Express.js, MongoDB, Socket.io, TailwindCSS

- Built oForge as a team collaboration platform for students, combining team discovery with a feature-rich workspace including Kanban boards, real-time chat, schema design, documentation, and whiteboard tools.
- Introduced a gated team collaboration system where users must submit proposals to join projects, with workspace access granted only after owner approval, ensuring privacy and controlled team formation.

KVMemo – Multithreaded In-Memory Key-Value Store demo
C++, TCP Sockets, Multithreading, Custom Thread Pool, Atomics

- Designed and implemented a high-performance in-memory key-value store in modern C++ supporting concurrent TCP client connections.
- Built a thread-pool architecture with task scheduling and synchronization using std::mutex, std::sharedmutex, and atomic counters to optimize read-heavy workloads.
- Implemented runtime metrics including QPS, latency tracking, hit/miss ratio, and active connection monitoring for performance observability.

CDS Library GitHub
C, C++

- Built modular header files for String, Stack, and Linked List with reusable C interfaces.
- Improved memory management and reduced redundancy using macros and pointer abstraction.

Education

Chitkara University, Chandigarh Oct 2022 – June 2026
B.E Computer Science *CGPA: 9.41*

Achievements

- Solved **1300+** DSA problems across LeetCode, GFG, and CodeChef & **LeetCode Contest Rating: 1618**.
- Selected for UCA Batch at Chitkara University – Top 3% out of 2500+ students.
- **Finalist** – Shopclues Smart AI Cataloging Hackathon (Top 5 of 70+ teams), Chandigarh, 2024.